NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L26.10 CFD Fault Injection and Repeated-Run Evaluation Benchmark

Group Rapid Trials Lab

Status Promoted

Period 2026-10-21 14:40 — 2026-10-26 16:40

Stage conclusion

Project report

L26.10 CFD fault-injection and repeated-run evaluation set

Current status

- 10-26 14:43 Work started. The J0 smoke job (clean baselines A and B, 1 run each, checking audit fields) was submitted and registered, with its experiment card. After reading J0, the lead froze the L1 check script and passed the K10 control. Jobs then run in series: J1 (baseline A + 14 injections, 18 runs), J2 (baseline B + 3 injections, 7 runs), J3 (R-b, 5 runs). Total: 32 runs.
- The 5 R-b setups that a bot generated from scratch are back. A static check found no imports and no file or network operations. All 5 case.json files match the task description (u=1, ρ=1, ν=1/22000, d=1, dt=0.05).
- 10-21 14:45 Launch plan written, awaiting PI approval. No computation before approval.
- The litwiki supplementary cards and the read-only review comments are back and merged into the plan (10-21 15:41).

Decisions made

- 10-26 14:43 Injection list finalized before the runs. The M6 fine-tier fault changed from "node count 1% lower than the source file" to "wrong mesh read (smoke instead of coarse)", which belongs to the K09 count-consistency class. The M4 convergence_pilot variant was dropped. The restart source is already bounded, so it stays bounded even if the pilot cannot be set; the injection is ineffective by construction. Run count 33→32.
- 10-26 14:43 The framework uses "observation mode": when L0 raises an error, it is logged and the run continues, so L1/L2 still score the same run. T28 itself would stop at that point. The PyFluent launch timeout was raised to 300 s, following Fluent's recommendation.
- 10-21 14:47 The clean baseline was split into A (cold start, for mesh/setup faults) and B (converged_segment restart, for numerical faults). Run count 27→33; diag_readonly does not count as L0.
- 10-21 Injection sites were split into an S1 implementation layer and an S2 specification layer. G2 only compares "readback against the manifest", so in principle it cannot catch specification-layer errors. This is the increment of this study over L26.6.
- 10-21 The L1 check script was written from the manifest text and frozen before the injection list was written (partially blind). The same lead designed the faults and wrote the script; the conclusion must state this.

Trials and results

Date	What was done	Result
10-26	J1, J2, J3 (30 runs, plus 2 in J0)	32 runs, 1.18 core-hours. Clean A and B: C_D bitwise identical across 5 runs each, zero false positives. All 17 injections took effect: L0 missed 8, L1 missed 0, L2 missed 12. R-b: 5/5 raised errors at the setup stage (all of the error-raising type) and were caught by L0
10-26	J0 smoke job: clean baselines A and B, 1 run each	Both finished with no L0 stop. Mean C_D over the last 20 steps: A 1.477, B 2.360. After 1 L1 calibration (A's dt now read from DT_READBACK), both runs passed L1/L2. K10 PASS. L1 frozen 10-26 14:52

Parameter sources

Parameter	Value	Source / assumption
Short-test steps and duration	200 steps, 144 s	Trials and results, 10-16 row
Square-cylinder setup	Re 2.2e4, u=d=ρ=1, ν=4.545e-5	
Plausible bluff-body C_D range	1–3	T28 item 3 (judgment of this line, not from literature)
Coarse-tier fault magnitudes	Actual values from the ledger	F02, F04–F08, C10, C21, C31
Fine tier = 1/5–1/10 of coarse tier; criterion thresholds 90% / 1/5 / 1/2 / 3× range	—	Assumption; no literature basis, based on this line's judgment

Open items and next steps

- 10-26 16:00 Draft conclusion sent to the coordinating lead. Deliverables for T28: a summary table, two suggested patch diffs (not tested in Fluent), and regression replay notes. The coordinating lead writes the handover file and decision record.
- Closure recommended (awaiting PI review): all criteria passed and the question is answered. Cluster cleanup: the run root totals 34 MB (runs 29 MB). All files used for verdicts are copied back to this line (14 MB) and can be recomputed in seconds. These are "run files of delivered cases"; under standing authorization the whole root is deleted 7 days after closure is approved, keeping this line's copy.

Literature basis

See launch plan §4 (10 litwiki cards added 10-21 15:41). The closest precedent is CFD Forge (arXiv:2610.14104, PDF p. 13). It planted errors in archived states; deterministic checks caught all of them, while a model that saw only the evidence passed 12/13. It did not rerun CFD.

PI notes

Citation snowballing (10-22, coordinating lead)

- At the PI's 10-22 request, the literature was expanded (OpenAlex one hop forward/backward + keywords), with a screening record.

- New papers not in the library and due for cards are listed in the screening record. The card request to litwiki is waiting to be sent (at 10-22 00:5x the platform-wide delegation queue was full; a retry is scheduled). This page's conclusions and criteria stay unchanged until the cards return. Possible effects are noted in the last section of the screening record.

Key points from new literature (10-22, after the cards returned, coordinating lead)

- Segura et al. 2016, survey of metamorphic testing: for numerical/PDE programs it only reports single fault cases detected by one metamorphic relation. It gives **no** overall mutation score for numerical programs. Limitations: deciding equivalent mutants, and constructing and selecting relations (accepted manuscript PDF pp. 12, 15–16; pp. 11, 14–15 in another copy). → A possible addition as "detectors without reference values" (e.g., near-zero mean lift under symmetric inflow, mirrored results after reversing the inlet direction). The literature gives no expected detection rate, so this is only an extra observation in this study and does not enter the criteria.
- Basiri et al. 2016, four principles of chaos engineering: form a hypothesis around steady-state behavior, inject real-world events, experiment in production, and automate continuously (author manuscript PDF pp. 4–8). → This has the same structure as this plan: 5 clean baseline repeats set the steady-state variation, then real faults from the ledger are injected. It can be cited as the method basis in the plan's literature section.

Key figures